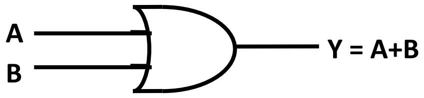
LOGIC GATES

Logic gate is an electronic circuit that performs a Boolean logical operation. A logic gate has one or more inputs but only one output.

- The basic gates are AND gate, OR gate and NOT gate.
- The universal gates are NAND gate and NOR gate.
- The special gates are XOR and XNOR.

OR gate:

OR gate is a basic gate which has two or more input but only one output. The output is high if any one of the input is high.



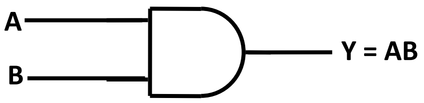
Logic symbol

Input		Output
A	B	**Y = A+B**
0	0	0
0	1	1
1	0	1
1	1	1

Truth table

AND gate:

AND gate is a basic gate which has two or more input but only one output. The output is high if all the inputs are high.



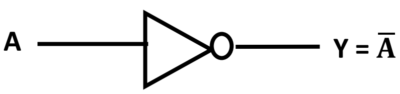
Logic symbol

Input		Output
A	B	**Y = AB**
0	0	0
0	1	0
1	0	0
1	1	1

Truth table

NOT gate:

It is a basic gate which complements the input signal value. It is also called as an inverter. It is a logic gate with only one input and one output.



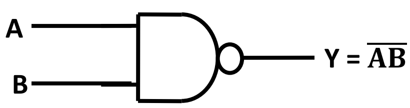
Logic symbol

Input	Output
A	**Y = $\overline{A}$**
0	1
1	0

Truth table

NAND gate:

NAND gate is a universal gate which performs complement of AND logic. NAND gate has two or more input but only one output. The output is high only when any of the input is low.



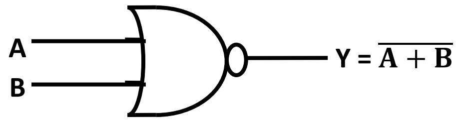
Logic symbol

Inputs		Output
A	B	**Y = $\overline{AB}$**
0	0	1
0	1	1
1	0	1
1	1	0

Truth table

NOR gate:

NOR gate is a universal gate which performs complement of OR logic. NOR gate has two or more input but only one output. The output is high if and only if all the inputs are low.



Logic symbol

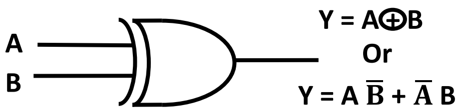
Inputs		Output
A	B	Y = $\overline{A + B}$
0	0	1
0	1	0
1	0	0
1	1	0

Truth table

Exclusive OR (XOR) gate:

XOR gate is a special gate which has two or more input but only one output. The output is high only when odd numbers of input are high.

XOR gate is also called as **Inequality detector**.



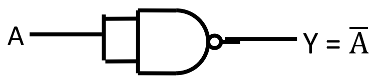
Logic symbol

Inputs		Output
A	B	Y = A $\overline{B}$ + $\overline{A}$ B
0	0	0
0	1	1
1	0	1
1	1	0

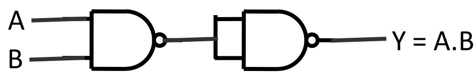
Truth table

Realization of NOT, AND, OR and XOR gates using NAND gate.

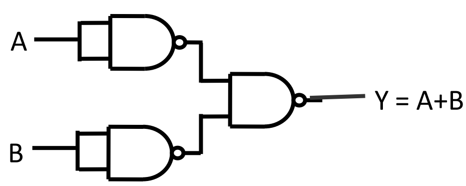
The NAND gate as a NOT gate



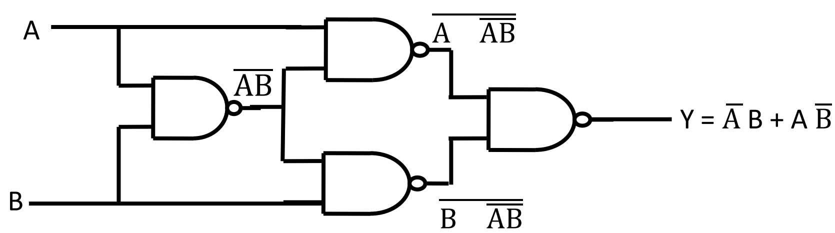
The NAND gate as an AND gate



The NAND gate as an OR gate

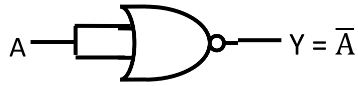


The NAND gate as an XOR gate

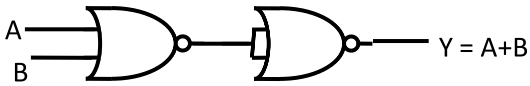


Realization of NOT, AND, OR and XNOR gates using NOR gate.

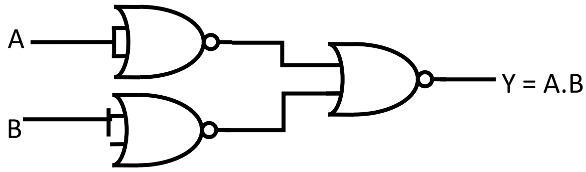
The NOR gate as a NOT gate



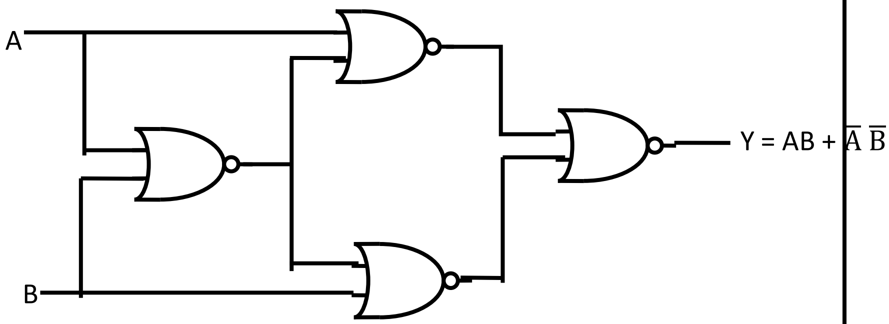
The NOR gate as a OR gate



The NOR gate as a AND gate



The NOR gate as a XNOR gate



Simplification of Boolean expressions

Product term: The logical product of Boolean variables, complemented or uncomplemented form is a product term.
Example: A $\overline{B}$ C, A B C, A B $\overline{C}$, A B, $\overline{A}$ $\overline{B}$ etc

Sum term: The logical sum of Boolean variables, complemented or uncomplemented form is a sum term.
Example: (A + $\overline{B}$ + C), (A + B), ($\overline{A}$ + $\overline{B}$ + C)

Sum of products (SOP): The logical sum of two or more logical product terms is known as sum of products.
Example: A $\overline{B}$ + $\overline{C}$ A + A B

Product of sum (POS): The logical product of two or more logical sum terms is known as product of sum.
Example: (A + $\overline{B}$) (B + C) (A + B)

Canonical SOP (or) Standard SOP expression: The Boolean expression containing all the input variables in each of the product term either in complemented or uncomplemented form is known as canonical SOP expression.

Canonical POS (or) Standard POS expression: The Boolean expression containing all the input variables in each of the sum term either in complemented or uncomplemented form is known as canonical POS expression.